

Immersive Visualization of Sneeze Simulation Data on Mobile Devices

Liangding Li*

University of Central Florida

Douglas Hector Fontes†

University of Central Florida.

Dirk Reiners‡

University of Central Florida.

Carsten Neumann‡

University of Central Florida.

Carolina Cruz-Neira||

University of Central Florida.

Michael Kinzel§

University of Central Florida.

ABSTRACT

One key factor in stopping the spread of COVID-19 is practicing social distancing. Visualizing possible sneeze droplets' transmission routes in front of an infected person might be an effective way to help people understand the importance of social distancing. This paper presents a mobile virtual reality (VR) interface that helps people visualize droplet dispersion from the target person's view. We implemented a VR application to visualize and interact with the sneeze simulation data immersively. Our application provides an easy way to communicate the correlation between social distance and infected droplets exposure, which is difficult to achieve in the real world.

Index Terms: Virtual Reality—Computational Fluid Dynamics(CFD)—COVID-19—Immersive visualization;

1 INTRODUCTION

COVID-19 infected people can spread the virus via small liquid particles from their mouth or nose when they cough, sneeze, speak, sing, or breathe. If infectious particles passing through the air are inhaled at a short distance, one can contact the virus and has the possibility to get infected. However, it is difficult to visualize the dispersion of sneeze droplets in real-world scenarios. With the help of recent technology, a visualization of simulation data can help people understand the effect more intuitively than from pure data or statements. Researchers at Florida Atlantic University [1] use laser light to show how far the particles can travel along their trajectory. They found that particles can travel up to 12 feet after a heavy cough. A web-based 3D simulation showing the importance of social distancing is also provided [2]. Compared with data visualization on 2D displays the immersive 3D visualization in a Virtual Reality (VR) environment can be more realistic and intuitive to comprehend for the user. Pavlou et al. [7] designed and developed a VR application based on COVID-19 indoor safety guidelines. Their application focused on simulating the user's behavior instead of the sneeze droplets' movement.

VR devices like head-mounted displays (HMDs) and CAVEs [4] provide people with an immersive way of interacting with data. Visualization can help people understand complex scientific data intuitively. An immersive visualization allows users to explore data from a first-person perspective and to interact with the data. Quam et al. [8] designed a system to visualize the blood flow data in vessels via CAVE. Coltekin et al. [3] discussed the usability and usefulness of geographic data in VR, and identified some of the central arguments for and against it.

*e-mail: leoriohope@knights.ucf.edu

†e-mail: douglashector.fontes@ucf.edu

‡e-mail: Carsten.Neumann@ucf.edu

§e-mail: michael.kinzel@ucf.edu

||e-mail: Dirk.Reiners@ucf.edu

||e-mail: Carolina@ucf.edu



Figure 1: Sneeze droplets hit simulated avatar in VR. The right-side avatar performs a sneeze. The user's perspective is the perspective of the left-side avatar in the application

The contribution of this work can be summarized as follows: We have designed a mobile VR interface that combines immersive visualization of the simulated sneeze data and user interface to interact with the data. In addition, we provided an easy way to reproduce the simulation in a VR environment. On top of that, we tested the correlation between social distancing and infected droplets exposure. Even though the CFD simulation is pre-computed, the immersive VR simulation can provide valuable educational information to the user.

2 DESIGN AND METHOD

The designed application includes the simulated sneeze data, the immersive computational fluid dynamics (CFD) visualization, and the interactive mobile user interface. All components are implemented and run in a smartphone application, which provides the user with an easy to use device.

2.1 Simulation

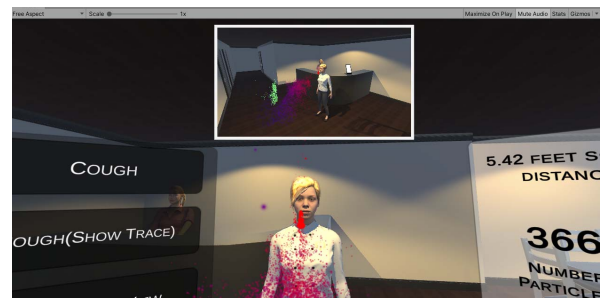
We use a simulation previously developed by Fontes et al. [5]. These simulations of a sneeze considers a human body within a room. The human body model was used to evaluate its influence on the dispersion of droplets during a sneeze. The human model, with a height of 1.78 meters, is located at the center of a cylindrical domain with 3 meters of radius and height. An inlet boundary condition drives the sneeze at the bottom surface of the throat of the human model. An artificial nasal passage is also included to improve the realism of the model. We did not consider droplet dynamics associated with droplet-droplet interactions (coalescence/stripping). The assumption that droplets do not interact with each other is valid in the dilute concentration regions with a low probability of collision [6] [9] exterior to the mouth. We included mass transfer between the droplets and the surrounding air through the evaporation/condensation model.

2.2 Visualization

The first part of the proposed application allows users to visualize droplets and their movement trace during a sneeze, as shown in Figure 1. We implemented the droplets visualization using Unity's particle system. The data set contains the position and velocity of



(a) Dynamic hits calculation



(b) Third person view to show the droplets movement trace

Figure 2: App environment and user interface

each particle in each step. First, we rendered all the droplets at the corresponding position for step i and treated them as stationary particles. In the next step, we kill the current particles at step i and render new particles based on the position at step $i + 1$.

In order to show the transmission trace of the droplets, we need to identify the same droplet at each time step. The simulation data has varying numbers of particles at each time step and does not use a consistent ordering between time steps. Therefore we reevaluated the data set and found the diameter of the particles can be used as a unique id across all time steps. We sorted the data set by particle diameter, made all time steps the same length, and added lifetime info to each row. This allows using Unity's build-in particle trails to show the droplets' movement paths.

To calculate the number of droplets that hit the user at a different distance, we created one sphere collider to represent the human head and another cylinder collider to represent the human body. When the particle enters the collider, a second particle system will be triggered and emits green particles to represent the droplets hitting the user. The number of hits will be recorded in real-time and shown on the UI panel, as shown in Figure 2a.

2.3 User Interface

The VR application was designed and implemented in the Unity game engine. The environment was set in a restaurant since the restaurant is a high-risk location for COVID-19 spread. The environment and the furniture models were designed and created by 3D model artists using Blender. We also created a sneeze animation and audio clips to apply to the avatars. In the application, the user acts as the person at potential risk of infection and stands in front of the infected person. To interact with the data on a smartphone (without the controller). We used a gaze timer to control the user interface (UI) buttons and trigger events. We assume the center of the screen as the gaze point, and when the user focuses on an interactive area for a while (we set the trigger time as 1.5 seconds), the corresponding event will trigger. For example, to trigger a sneeze event, the user needs to lock the target avatar's gaze position for 1.5 seconds. The UI menu will be activated at the same time. We allow the user to show/hide the droplet trace, and we added a second view from the side to visualize particle hits on the user at a different angle (Figure 2b). To change the distance between the user and the target avatar, the user can gaze at the move front/back buttons on the UI panel.

3 RESULTS

We run the application on a mobile device with Android OS. The application can run an average at 50fps on the target device. We simulated the sneeze in the virtual environment at distances of 2ft, 4ft, 6ft, 8ft and recorded the number of hits. We repeated the simulation at each distance five times and calculated the average hits. Since we used sphere and cylinder colliders to represent the user's body, the

user's head movement and rotation will not affect the contact area. Users can only move forward and back. Free movement is disabled.

4 CONCLUSION AND FUTURE WORK

In this work, we provided an easy and cost-effective way to visualize sneeze simulation data in an immersive virtual environment, which helps non-professionals understand the importance of social distancing during COVID-19. Even though the simulation and distance test in VR is not as accurate as performing an offline CFD simulation, the user can still get an intuitive feeling about the correlation between distance and number of droplet impacts during a sneeze.

We only use the sneeze simulation data set at this stage. To improve the application's versatility in the future, we plan to simulate various coughs and sneezes, including some heavy coughs. We also plan to include simulations with the infected person wearing a mask. By comparing the hits between with/without mask situations, we can evaluate the mask's ability to reduce droplets' dispersion.

REFERENCES

- [1] STAY 6 FEET APART. SIMULATED COUGH REVEALS THAT MAY NOT BE ENOUGH. 2020. To appear.
- [2] This 3-D Simulation Shows Why Social Distancing Is So Important. 2020. To appear.
- [3] A. Çöltekin, I. Lokka, M. Zahner, and L. Halounova. On the usability and usefulness of 3d (geo) visualizations—a focus on virtual reality environments. *ISPRS*, 2016.
- [4] C. Cruz-Neira, D. J. Sandin, and T. A. DeFanti. Surround-screen projection-based virtual reality: the design and implementation of the cave. In *Proceedings of the 20th annual conference on Computer graphics and interactive techniques*, pp. 135–142, 1993.
- [5] D. Fontes, J. Reyes, K. Ahmed, and M. Kinzel. A study of fluid dynamics and human physiology factors driving droplet dispersion from a human sneeze. *Physics of Fluids*, 32(11):111904, 2020.
- [6] C. Lun and H. Liu. Numerical simulation of dilute turbulent gas-solid flows in horizontal channels. *International Journal of Multiphase Flow*, 23(3):575–605, 1997.
- [7] M. Pavlou, K. Kalatzis, A. Chrysanthakopoulou, D. Laskos, S. Georgakopoulos, D. Voultsidis, and K. Moustakas. Remote adversarial vr serious game simulating covid-19 infection spread and protection protocols. In *2021 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 683–684. IEEE, 2021.
- [8] D. J. Quam, T. J. Gundert, L. Ellwein, C. E. Larkee, P. Hayden, R. Q. Migrino, H. Otake, and J. F. LaDisa Jr. Immersive visualization for enhanced computational fluid dynamics analysis. *Journal of biomechanical engineering*, 137(3):031004, 2015.
- [9] D. Sun. Four-way coupled second-order method of two-fluid model for gas-particle flow and the numerical simulation in horizontal channels. *Powder Technology*, 390:354–368, 2021.